Software Migration Module (SMM)

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OOF® Origin Open Foundation®

Independent Methodological Authority

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Canonical Definition

Software Migration Module (SMM) is the Core Module within Software Retirement Standard (SRTS) that governs how software functions, operational capabilities, users, data relationships, configurations, interfaces, and dependent activities are transferred, replaced, reconstructed, or deliberately discontinued when an identifiable software object is being retired.

SMM establishes software migration as an evidence-supported governance

transition between a defined source condition and an intended destination condition rather than merely a technical transfer of software artifacts, data, or workloads.

The module distinguishes migration planning from migration authorization,

technical transfer from functional continuity, destination availability from destination suitability, migration execution from migration verification, and successful migration from retirement closure.

It requires material migration activities to remain associated with the

source software, intended destination, migration purpose, affected functions, dependencies, applicable authority, transformation rules, execution evidence, verification criteria, and resulting operational state.

SMM recognizes that migration may involve direct replacement, platform transition, version transition, service consolidation, architectural transformation, data transfer, workflow reassignment, or redistribution of software capabilities across multiple destination systems.

It also recognizes that not every function or data object must be preserved unchanged, provided that any material discontinuation, transformation, exclusion, or loss is explicitly justified and governed.

The module governs migration conditions and outcomes without assuming that the retiring software can be withdrawn merely because the destination software has become operational.

SMM applies to enterprise applications, cloud services, software platforms,

databases, APIs, embedded systems, AI services, autonomous agents, and software operating within industrial or critical infrastructure environments.

Its principal governance output is the Software Migration Governance Profile.

Operational Role

SMM establishes the governance requirements for controlled software

migration within the retirement lifecycle.

It governs:

• Migration case identity

• Migration origin and purpose

• Source software identification

• Destination software identification

• Migration scope and boundaries

• Current and intended operational conditions

• Migration suitability

• Functional continuity

• Data continuity

• User continuity

• Configuration transition

• Interface transition

• Dependency transition

• Operational responsibility transfer

• Migration method selection

• Migration transformation rules

• Migration sequencing

• Migration prerequisites

• Migration readiness

• Migration authority

• Migration execution traceability

• Parallel operation

• Cutover conditions

• Migration verification

• Migration deviations

• Recovery and fallback

• Partial migration

• Residual source dependencies

• Migration outcome classification

• Migration closure

• Handoff to residual dependency and retirement closure governance

SMM does not independently authorize the retirement of source software or establish that all retirement obligations have been satisfied.

Operational Question

How can software capabilities, data, users, interfaces, dependencies, and operational responsibilities be transferred or deliberately transformed from a retiring software environment to an intended destination while preserving justified continuity, applicable authority, verifiable outcomes, and visibility into everything that remains unresolved?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting: Software Lifecycle Control

SRTS Core Module Flow

Retirement Planning Module (RPM) → End-of-Support Management Module (ESMM)

→ Software Migration Module (SMM) → Residual Dependency Management Module (RDMM) → Retirement Closure Control Module (RCCLM)

SMM is the third Core Module within SRTS.

It governs the transition requirements established through retirement planning and affected by support conditions.

It provides migration outcomes and evidence to residual dependency

management and retirement closure.

Migration may proceed in phases and may require repeated interaction with other SLGA™ lifecycle mechanisms.

• Governed Space

Software Migration Governance

This governed space concerns the controlled transition of software-related

functions, data, users, interfaces, dependencies, and operational responsibilities between identifiable source and destination conditions.

It begins when migration becomes a required or selected part of a software retirement pathway.

It includes migration scope, source and destination identity, suitability,

transformation requirements, continuity conditions, readiness, authorization, execution, verification, and unresolved consequences.

It remains applicable where migration is incomplete, partially verified, reversed, or subject to continuing transition obligations.

It concludes for a defined migration case when its actual outcome has been established and remaining migration obligations have been appropriately resolved or transferred.

Governed Object

The primary Governed Object is:

A Software Migration Case representing the evidence-supported relationship

between an identifiable source software condition, an intended destination condition, the software capabilities and relationships subject to transition, applicable migration requirements, authorized execution activities, verification evidence, resulting operational state, and unresolved migration obligations.

The governed object must preserve, where applicable:

Migration Case ID

Retirement Case Reference

Source Software ID

Source Version and Configuration

Source Operational State

Destination Software ID

Destination Version and Configuration

Intended Destination State

Migration Purpose

Migration Scope

Affected Functions

Affected Data

Affected Users

Affected Interfaces

Affected Dependencies

Transformation Requirements

Migration Authority

Migration Method

Migration Status

Execution Evidence

Verification Evidence

Residual Obligations

Migration Record Version

Scope

SMM governs:

Software Replacement Migration

Software Version Migration

Platform Migration

Application Migration

Service Migration

Cloud Migration

Data-Associated Software Migration

Interface Migration

Workflow Migration

Configuration Transition

Dependency Transition

AI Service Migration

Autonomous Agent Capability Transfer

Embedded Software Transition

Partial Migration

Phased Migration

Parallel Operation

Migration Cutover

Migration Verification

Migration Recovery

Migration Closure

SMM governs migration relationships relevant to software retirement.

It does not replace the specialized governance of software creation, configuration, security, release, deployment, or data management.

Foundational Principle

Software migration is successful only when the intended destination condition and required continuity of functions, data, dependencies, and operational responsibilities have been established through applicable evidence—not merely when a technical transfer has completed.

Migration must be evaluated against what the source software actually provided and what the destination is legitimately required to preserve, replace, or discontinue.

Transfer Completed ≠ Migration Verified.

Destination Operational ≠ Source Retirement Authorized.

Migration Case Identity

Every material migration must have a stable identity.

Relevant elements include:

Migration Case ID

Retirement Case Reference

Source Software Reference

Destination Software Reference

Migration Origin

Responsible Function

Creation Time

Migration Status

Evidence Reference

Record Version

The identity must remain stable across planning, execution, verification, recovery, and closure.

Migration Origin

Migration may originate from:

Software Retirement Requirement

Support Termination

Security Exposure

Technology Obsolescence

Infrastructure Transition

Software Consolidation

Platform Replacement

Dependency Discontinuation

Operational Modernization

External Requirement

Autonomous System Replacement

The migration origin must remain distinguishable from migration authorization.

Migration Purpose

Every migration case must establish its intended purpose.

Potential purposes include:

Preserve Required Functionality

Replace Unsupported Software

Maintain Data Accessibility

Transfer Operational Workflows

Remove Legacy Dependencies

Improve Security Conditions

Enable Software Retirement

Consolidate Software Services

Transfer Autonomous Responsibilities

Complete Platform Transition

The purpose must identify the intended operational outcome rather than only the transfer mechanism.

Source Software Identification

SMM must establish the identity of the source software.

Relevant elements include:

Software Object ID

Product or Service Name

Version

Build

Artifact Reference

Configuration

Deployment Environment

Operational Function

Support Status

Dependency Relationships

The source identity must be sufficiently precise to distinguish affected instances from software outside the migration scope.

Source Operational State

SMM must establish the actual source condition.

Relevant elements include:

Active Functions

Users

Data Relationships

Interfaces

Scheduled Activities

Configurations

Dependencies

Security Permissions

Operational Restrictions

Recovery Capabilities

The migration basis must not rely solely on incomplete documentation or assumed source behavior.

Destination Software Identification

SMM must identify the intended destination.

Relevant elements include:

Destination Software ID

Product or Service Name

Version

Build

Platform

Configuration

Deployment Environment

Operational Owner

Support Conditions

Applicable Governance References

A destination may consist of multiple software objects rather than one replacement application.

Destination Operational State

SMM must define the intended destination condition.

Relevant elements include:

Required Functions

Required Users

Required Data

Required Interfaces

Required Configurations

Required Dependencies

Required Security Conditions

Required Performance

Required Availability

Required Recovery Capability

The destination condition must be defined sufficiently to support verification.

Source–Destination Relationship

SMM must preserve the relationship between source and destination objects.

Potential relationships include:

One-to-One Replacement

One-to-Many Distribution

Many-to-One Consolidation

Many-to-Many Transformation

Version-to-Version Transition

Platform-to-Platform Transition

Service-to-Service Transition

Software-to-Manual Process Transition

Capability Discontinuation

The relationship must reflect the actual migration structure.

Migration Scope

Migration scope must identify what is intended to move, change, remain, or cease.

Relevant dimensions include:

Software Functions

Data Objects

Users

Interfaces

Configurations

Dependencies

Operational Workflows

Scheduled Activities

Security Relationships

Monitoring and Recovery Functions

The scope must distinguish complete migration from partial migration.

Migration Boundary

SMM must establish the boundaries of the migration case.

Relevant boundaries include:

Source Boundary

Destination Boundary

Version Boundary

Environment Boundary

Data Boundary

Interface Boundary

Dependency Boundary

Authority Boundary

Operational Responsibility Boundary

Verification Boundary

Excluded objects must remain visible where they are materially affected.

Migration Object Classification

Migration objects may be classified as:

Preserved

Transferred

Transformed

Reconstructed

Replaced

Consolidated

Distributed

Archived

Deliberately Discontinued

Excluded

Unresolved

Unknown

Each material classification must preserve its justification and applicable evidence.

Functional Continuity

SMM must identify functions that must continue after migration.

Relevant elements include:

Function Identity

Source Behavior

Required Destination Behavior

Users

Inputs

Outputs

Dependencies

Operational Conditions

Verification Criteria

Functional continuity does not necessarily require identical implementation.

It requires satisfaction of the applicable functional obligations.

Functional Equivalence

Where functional equivalence is required, SMM must define the relevant criteria.

Potential dimensions include:

Input Compatibility

Output Compatibility

Behavioral Consistency

Business Rule Preservation

Error Handling

Operational Reliability

Timing Requirements

Security Requirements

External Interface Behavior

Equivalence must be evaluated against the actual governed requirements.

Deliberate Functional Discontinuation

Some functions may be intentionally removed during migration.

SMM must preserve:

Function Identity

Discontinuation Purpose

Affected Users

Operational Consequences

Applicable Authority

Dependency Implications

Communication Requirements

Verification Evidence

A function must not disappear from the migration record merely because the destination does not implement it.

Data Migration Relationship

SMM must identify data affected by migration.

Relevant elements include:

Source Data Object

Destination Data Object

Data Ownership

Data Structure

Transformation Requirements

Integrity Requirements

Accessibility Requirements

Retention Requirements

Verification Method

Detailed data governance remains subject to applicable specialized

mechanisms.

Data Mapping

Where data is transferred or transformed, SMM must preserve the applicable mapping.

Relevant elements include:

Source Field or Object

Destination Field or Object

Transformation Rule

Required Validation

Known Incompatibility

Excluded Data

Loss or Precision Limitation

Mapping Version

Evidence

A successful import does not establish that the intended meaning of the data has been preserved.

Data Integrity

SMM must identify the integrity conditions required for migration.

Relevant elements include:

Completeness

Consistency

Accuracy

Relationship Preservation

Identity Preservation

Temporal Integrity

Transformation Validity

Duplicate Handling

Error Classification

Integrity verification must be appropriate to the actual data and migration requirements.

Data Accessibility

Migration must address continued access to data where required.

Potential conditions include:

Active Data Access

Historical Data Access

Archived Data Access

Read-Only Access

Recovery Access

External Reporting Access

Regulatory Access

User Access

Data accessibility requirements must remain distinct from data transfer completion.

Data Retention and Disposal

SMM must identify migration-related retention and disposal requirements.

Relevant elements include:

Source Data Retention

Destination Data Retention

Archive Requirements

Backup Retention

Deletion Conditions

Historical Evidence

Responsible Function

Applicable Authority

Migration does not independently authorize deletion of source data.

User Continuity

SMM must identify users affected by migration.

Relevant elements include:

User Groups

Access Requirements

Identity Continuity

Role Mapping

Authentication Changes

Authorization Changes

Required Training

Communication Requirements

Operational Restrictions

User continuity must reflect actual access and capability requirements.

Identity and Permission Transition

Where migration changes software identities or permissions, SMM must identify:

Source Identity

Destination Identity

User Identity Mapping

Service Identity Mapping

Access Rights

Delegated Permissions

Credential Transition

Revocation Requirements

Authority Verification

Migration must not create unapproved expansion of authority.

Configuration Transition

SMM must identify configuration relationships affected by migration.

Relevant elements include:

Source Configuration

Destination Configuration

Configuration Mapping

Environment Variables

Feature Settings

Security Settings

External Connections

Configuration Dependencies

Verification Evidence

Configuration transition must remain subject to applicable configuration governance.

Interface Migration

SMM must identify interfaces requiring transition.

Relevant elements include:

Source Interface

Destination Interface

Dependent Consumers

Protocol Compatibility

Data Format Compatibility

Authentication Requirements

Version Compatibility

Cutover Conditions

Verification Evidence

A destination interface must not be considered equivalent merely because it exposes a similar name or endpoint.

Dependency Transition

SMM must identify dependencies that must be replaced, transferred, retained, or removed.

Relevant categories include:

Software Libraries

Runtime Platforms

Databases

External APIs

Infrastructure Services

Authentication Systems

Monitoring Systems

Operational Processes

Physical-System Dependencies

The migration must preserve visibility into dependencies that remain attached to the source software.

Upstream Dependency Transition

Upstream dependencies are resources required by the source or destination software.

SMM must establish:

Source Dependency

Destination Dependency

Replacement Requirement

Compatibility Condition

Support Status

Security Condition

Transition Evidence

Residual Relationship

Shared dependencies must not be removed solely because the source software is being retired.

Downstream Consumer Transition

Downstream consumers rely on the software being migrated.

Relevant elements include:

Consumer Identity

Consumed Function or Interface

Migration Requirement

Destination Compatibility

Transition Timing

Verification Evidence

Unresolved Consumer

Responsible Function

Unknown or unverified consumers must remain explicit migration uncertainty.

Operational Workflow Migration

SMM must identify workflows affected by migration.

Relevant elements include:

Workflow Identity

Source Execution Path

Destination Execution Path

Required Inputs

Expected Outputs

Human Responsibilities

Automated Activities

Exception Handling

Verification Conditions

Workflow migration must preserve required operational outcomes, not merely technical connectivity.

Autonomous Agent Migration

Where autonomous software is replaced or migrated, SMM must assess:

Agent Identity

Assigned Objectives

Active Tasks

Delegated Authority

Tool Permissions

Memory References

Scheduled Activities

External Commitments

Successor Agent

Human Override

Transfer of tasks must remain distinguishable from transfer of authority.

Agent Authority Transfer

SMM must identify whether delegated authority is being terminated, reissued, or transferred.

Relevant elements include:

Source Agent Authority

Successor Identity

Authorized Objectives

Tool Permissions

External System Access

Delegation Conditions

Revocation Requirements

Authorization Evidence

Effective Authority State

The existence of a successor agent does not automatically authorize it to inherit the predecessor's permissions.

Task Continuity ≠ Authority Continuity.

AI Model and Service Migration

Where AI services are migrated, SMM must identify:

Source Model or Service

Destination Model or Service

Version Relationship

Interface Compatibility

Input and Output Differences

Behavioral Requirements

Safety Requirements

Performance Conditions

Dependent Applications

Verification Evidence

Model availability does not establish behavioral or operational equivalence.

Embedded Software Migration

Where software operates within physical systems, SMM must assess:

Device Identity

Source Firmware

Destination Firmware

Hardware Compatibility

Operational Function

Physical Safety Conditions

Recovery Capability

Installation Conditions

Verification Requirements

Software migration must not assume that a technically installable version is operationally suitable for the physical system.

Migration Method Selection

SMM must identify the selected migration method and its justification.

Potential methods include:

Direct Replacement

Phased Migration

Parallel Operation

Incremental Transfer

Batch Migration

Rolling Transition

Service Consolidation

Platform Rebuild

Controlled Capability Discontinuation

The selected method must reflect the actual migration scope, dependencies, risks, and continuity requirements.

Migration Method Suitability

Migration method suitability must consider:

Operational Criticality

Data Volume and Complexity

Dependency Relationships

Compatibility Requirements

Downtime Constraints

Verification Capability

Recovery Feasibility

Resource Availability

Security Requirements

Method selection must not be based solely on execution speed.

Migration Sequencing

SMM must establish the sequence of material migration activities.

A possible sequence is:

Source Identification → Destination Readiness → Dependency Preparation →

Data and Function Transition → Interface Transition → Operational Verification → Cutover → Residual Source Assessment

Sequencing must reflect prerequisite relationships and actual operational conditions.

Migration Preconditions

SMM must identify conditions required before migration activities proceed.

Potential preconditions include:

Applicable Migration Decision

Source State Established

Destination State Defined

Required Dependencies Available

Data Mapping Established

Security Requirements Addressed

Operational Resources Available

Verification Capability Available

Recovery Conditions Prepared

Competent Authority Established

Unmet mandatory preconditions must remain explicit.

Migration Readiness

Migration readiness must be assessed against the actual migration scope.

Relevant elements include:

Source Inventory Verified

Destination Available

Functional Requirements Established

Data Requirements Established

Dependencies Assessed

Interfaces Prepared

Authority Established

Execution Resources Available

Verification Criteria Defined

Fallback Conditions Established

Readiness does not establish migration completion.

Migration Authority

SMM must identify authority for material migration decisions and actions.

Relevant elements include:

Migration Decision Authority

Source Software Authority

Destination Software Authority

Data Transfer Authority

Configuration Change Authority

Deployment Authority

Access Change Authority

Cutover Authority

Recovery Authority

External Approval Requirements

Authorization of the migration objective does not automatically authorize every constituent action.

Migration Responsibility

SMM must identify responsible functions.

Relevant elements include:

Migration Owner

Source System Owner

Destination System Owner

Data Responsible Function

Security Responsible Function

Infrastructure Responsible Function

Verification Responsible Function

Operational Transition Responsible Function

Recovery Responsible Function

Responsibilities must remain aligned with applicable authority and capability.

Migration Execution Traceability

Material migration activities must remain traceable.

Relevant elements include:

Execution Activity ID

Migration Case Reference

Source Object

Destination Object

Responsible Actor

Applicable Authority

Execution Time

Execution Method

Observed Result

Evidence Reference

Traceability must support reconstruction of what was transferred, transformed, excluded, or left unresolved.

Automated Migration

Automated migration tools may execute authorized transition activities.

SMM must preserve:

Automation Identity

Assigned Task

Execution Scope

Applicable Permissions

Source Access

Destination Access

Transformation Rules

Execution Evidence

Exception Handling

Human Oversight Requirements

Automation must not infer additional authority from technical access.

Parallel Operation

Some migrations require source and destination systems to operate

simultaneously.

Relevant elements include:

Parallel Operation Purpose

Source Role

Destination Role

Data Synchronization

Consistency Requirements

User Routing

Conflict Handling

Monitoring Requirements

Exit Conditions

Parallel operation must not create uncontrolled divergence between software states.

Data Synchronization

Where source and destination systems exchange data during migration, SMM must identify:

Synchronization Direction

Authoritative Data Source

Update Rules

Conflict Resolution

Latency Requirements

Consistency Requirements

Failure Handling

Verification Evidence

The existence of synchronized data does not automatically establish semantic or operational equivalence.

Migration Cutover

Cutover is the governed transition of operational reliance from source to destination.

Relevant elements include:

Cutover Scope

Cutover Preconditions

Cutover Authority

Operational Timing

Affected Users

Dependent Systems

Verification Requirements

Fallback Conditions

Cutover Evidence

Cutover must remain distinct from the retirement closure of the source software.

Cutover Verification

SMM must establish whether the intended operational transition has occurred.

Relevant elements include:

Destination Operational State

Required Functions Available

Required Data Accessible

Users Transitioned

Interfaces Operational

Dependencies Functional

Security Conditions Established

Source Reliance Reduced or Terminated as Intended

Verification Evidence

Successful cutover does not automatically establish that every source dependency has been eliminated.

Migration Verification

Migration verification must evaluate the actual result against the intended destination condition.

Relevant dimensions include:

Functional Verification

Data Integrity Verification

Interface Verification

Dependency Verification

Configuration Verification

Security Verification

Operational Continuity Verification

Performance Verification

Recovery Verification

Verification must be proportionate to the migration's actual significance and obligations.

Verification Evidence Applicability

Migration evidence must remain associated with:

Source Software Identity

Destination Software Identity

Version

Configuration

Environment

Migration Scope

Execution Time

Verification Method

Observed Result

Known Limitations

Evidence from one migration environment must not automatically establish successful migration in another.

Migration Outcome Assessment

SMM must compare the intended and actual migration outcomes.

Relevant elements include:

Intended Destination State

Actual Destination State

Transferred Functions

Transferred Data

Transitioned Users

Transitioned Interfaces

Resolved Dependencies

Unresolved Dependencies

Known Deviations

Residual Obligations

Outcome assessment must distinguish technical execution completion from verified operational success.

Partial Migration

Migration may be completed for only part of the intended scope.

Relevant elements include:

Completed Objects

Unmigrated Objects

Partially Migrated Functions

Unverified Data

Remaining Users

Residual Interfaces

Outstanding Dependencies

Required Follow-up

Partial migration must not be represented as complete migration.

Migration Deviation

A migration deviation occurs when actual activities or outcomes materially differ from the established migration conditions.

Potential deviations include:

Unexpected Data Transformation

Missing Functionality

Unplanned Interface Change

Incomplete User Transition

Dependency Failure

Unauthorized Configuration Change

Security Condition Failure

Unexpected Source Execution

Material deviations must trigger appropriate reassessment or corrective governance.

Migration Failure

Migration failure may occur when:

Destination Software Is Unavailable

Required Functions Fail

Data Integrity Cannot Be Established

Critical Interfaces Are Incompatible

Operational Continuity Is Compromised

Required Authority Is Absent

Security Requirements Are Not Satisfied

Verification Cannot Be Completed

Failure must remain traceable to the actual migration state and required response.

Migration Recovery

Where recovery is required, SMM must preserve:

Failure Condition

Affected Software

Recovery Objective

Fallback Pathway

Data Recovery Requirements

Operational Restoration Conditions

Recovery Authority

Recovery Execution Evidence

Resulting State

Recovery must not assume that returning to the source software automatically restores the original operational condition.

Migration Reversibility

SMM must assess whether material migration activities can be reversed.

Potential classifications include:

Fully Reversible

Conditionally Reversible

Partially Reversible

Operationally Irreversible

Unknown

Relevant considerations include:

Data Transformation

Schema Changes

Source Modification

Credential Changes

Interface Withdrawal

External System Effects

User Activity During Migration

Reversibility must be evaluated against the actual resulting state.

Residual Source Dependencies

After migration, the source software may remain necessary for specific functions or obligations.

Potential residual dependencies include:

Historical Data Access

Legacy Interface Consumers

Recovery Functions

Unmigrated Workflows

External Integrations

Archived Execution Requirements

Untransferred Automated Tasks

Physical-System Dependencies

These dependencies must remain identifiable and be handed to RDMM where applicable.

Residual Migration Obligations

Migration may leave unresolved obligations.

Relevant elements include:

Unverified Functions

Unmigrated Data

Incomplete User Transition

Outstanding Interface Changes

Unresolved Security Conditions

Incomplete Dependency Transfer

Pending Authority Revocation

Required Documentation

Post-Migration Monitoring

Residual obligations must remain assigned and traceable.

Migration Monitoring

Some migration outcomes require continued observation.

Potential monitoring concerns include:

Destination Stability

Data Consistency

Interface Errors

Unexpected Source Use

Performance Changes

Security Events

User Access Problems

Dependency Failures

Recovery Readiness

Monitoring must reflect the actual migration risks and operational significance.

Migration Reassessment

SMM must support reassessment when:

Source State Changes

Destination State Changes

Migration Scope Changes

New Dependencies Are Discovered

Data Requirements Change

Security Conditions Change

Verification Fails

Operational Continuity Is Threatened

Replacement Suitability Changes

Reassessment must preserve the earlier migration plan, decisions, and evidence.

Migration Classification

SMM recognizes the following governance classifications:

Migration Proposed

Migration Under Assessment

Migration Planned

Migration Readiness Pending

Migration Ready

Migration Authorized

Migration Active

Parallel Operation Active

Cutover Pending

Cutover Completed

Verification Pending

Migration Verified

Partially Migrated

Migration Completed With Residual Obligations

Migration Blocked

Migration Failed

Recovery Active

Migration Reversed

Migration Deferred

Reassessment Required

Migration Closed

Unknown

Conflicted

Each classification must remain associated with its applicable scope, evidence, and time.

Migration Closure

Migration closure is the evidence-supported determination of the disposition of an identifiable migration case.

Closure must preserve:

Migration Case Identity

Source Software

Destination Software

Intended Migration Outcome

Actual Migration Outcome

Verification Evidence

Known Deviations

Residual Source Dependencies

Outstanding Obligations

Closure Authority

Closure Decision

A migration case may be closed under an appropriately qualified disposition while residual obligations remain governed elsewhere.

Migration closure must not be treated as software retirement closure.

Governance Requirements

A conforming SMM implementation must satisfy the following requirements. Migration Identity and Scope

• Establish a stable migration case identity

• Identify the source software

- Identify the destination software or intended replacement condition
- Establish migration purpose
- Define migration scope
- Establish migration boundaries
- Identify the current source state
- Define the intended destination state

Continuity and Transformation

- Identify required functional continuity
- Establish applicable functional equivalence criteria
- Identify deliberate functional discontinuations
- Identify affected data relationships
- Establish data mapping where applicable
- Establish data integrity requirements
- Identify data accessibility obligations
- Identify affected users
- Identify identity and permission transitions
- Identify configuration transition requirements
- Identify interface transition requirements
- Identify upstream and downstream dependencies

Method and Readiness

- Establish the migration method
- Evaluate method suitability
- Establish migration sequencing
- Identify mandatory preconditions
- Establish migration readiness criteria
- Identify required resources and capabilities
- Identify competent migration authority
- Identify responsible functions
- Establish verification criteria
- Establish recovery and fallback requirements

Execution and Verification

- Preserve material execution traceability
- Govern automated migration within applicable authority
- Establish parallel operation controls where applicable

• Establish cutover conditions

• Verify material migration outcomes

• Preserve verification evidence

• Identify deviations and failures

• Preserve recovery results

• Assess actual destination conditions

• Prevent technical transfer completion from substituting for

migration verification.

Residual Obligations and Closure

• Identify residual source dependencies

• Identify unresolved migration obligations

• Establish post-migration monitoring where required

• Preserve reassessment triggers

• Classify migration outcomes

• Establish migration closure eligibility

• Preserve competent closure authority

• Maintain continuity with retirement planning

• Transfer relevant residual dependencies to RDMM

• Prevent migration closure from substituting for retirement closure

Governance Logic

Migration Origin → Migration Case Identity → Source Software

Identification → Source Operational State → Destination Identification → Intended Destination State → Migration Purpose → Migration Scope → Functional Requirements → Data Requirements → User and Interface Requirements → Dependency Assessment → Transformation Rules → Migration Method → Migration Preconditions → Migration Readiness → Applicable Authority → Governed Execution → Cutover Where Applicable → Verification → Actual Destination State → Residual Source Dependencies → Migration Outcome Classification → Migration Closure

Where migration is direct:

Source Software → Destination Readiness → Authorized Transfer →

Verification → Operational Transition

Where migration is phased:

Source Software → Migration Phases → Intermediate States → Phase

Verification → Final Transition

Where parallel operation is required:

Source and Destination → Parallel Operation → Synchronization →

Consistency Verification → Cutover

Where migration fails:

Migration Activity → Failure Recognition → Actual State Assessment →

Recovery / Reassessment → Revised Migration Pathway

Where source dependencies remain:

Migration Verified → Residual Source Dependency Identification → RDMM

Handoff → Retirement Closure Assessment

SMM must not equate destination operation with retirement of the source software.

Governance Outputs

The principal output is:

Software Migration Governance Profile

Supporting outputs may include:

Software Migration Case Identity

Source Software State Profile

Destination Software State Profile

Source–Destination Relationship Map

Migration Scope Record

Migration Object Classification

Functional Continuity Requirements

Functional Equivalence Assessment

Data Migration Mapping

Data Integrity Verification

User Continuity Profile

Identity and Permission Transition Record

Configuration Transition Record

Interface Migration Profile

Dependency Transition Map

Migration Method Assessment

Migration Readiness Profile

Migration Authority Record

Migration Execution Record

Cutover Verification Record

Migration Outcome Assessment

Migration Deviation Register

Migration Recovery Record

Residual Source Dependency Register

Migration Closure Record

Minimum Implementation Framework

A minimum SMM implementation must establish eight governance layers. Layer 1 — Migration Identity

Identify the source software, intended destination, migration purpose, and

retirement relationship. Layer 2 — Scope and Continuity

Define affected functions, data, users, interfaces, dependencies, and

operational obligations. Layer 3 — Source–Destination Mapping

Establish what must be preserved, transferred, transformed, replaced, or deliberately discontinued. Layer 4 — Migration Method and Preconditions

Select a justified migration method, establish sequencing, and identify

mandatory prerequisites. Layer 5 — Authority and Readiness

Establish responsible actors, applicable authority, execution capability,

verification requirements, and recovery conditions. Layer 6 — Governed Execution

Preserve traceability of material migration activities, automated actions,

parallel operation, and cutover. Layer 7 — Outcome Verification

Compare the intended destination state with the actual result and identify

deviations, failures, and uncertainty. Layer 8 — Residual Obligations and Closure

Preserve unresolved source dependencies, outstanding migration obligations,

closure disposition, and handoff to subsequent retirement governance.

The framework may be implemented through manual processes, migration platforms, software delivery pipelines, infrastructure orchestration, data migration tools, or AI-assisted systems.

Implementation technology must remain subordinate to the governed migration requirements.

Cross-Module Boundary

RPM → SMM

RPM establishes retirement planning requirements, migration necessity,

initial feasibility, and intended transition conditions.

SMM governs the detailed migration relationship, execution traceability,

verification, and outcome.

ESMM → SMM

ESMM establishes support and maintenance conditions that may constrain migration timing, capability, or technical assistance.

SMM governs migration under those applicable conditions.

SMM → RDMM

SMM identifies source dependencies and obligations that remain after migration.

RDMM governs their subsequent classification, responsibility, restriction,

transfer, and resolution.

SMM → RCCLM

SMM establishes the verified migration outcome and unresolved migration

obligations.

RCCLM evaluates their significance for software retirement closure.

Migration establishes the destination transition. It does not independently establish the final retirement state of the source software.

Cross-Architecture Boundary

SMM governs migration relationships within SRTS.

It does not replace specialized SLGA™ lifecycle governance.

SRS — Software Requirements Standard

Governs functional and nonfunctional requirements for destination software.

SCS — Software Creation Standard

Governs creation of replacement software, migration tools, and modified

components.

SDS — Software Dependency Standard

Governs dependency identity, suitability, change, and continuity.

SCFS — Software Configuration Standard

Governs source and destination configuration, artifact identity, and

configuration changes.

SQS — Software Quality Standard

Governs quality criteria, testing, defects, and condition assessment.

SSS — Software Security Standard

Governs security requirements, testing, vulnerabilities, and remediation.

SRLS — Software Release Standard

Governs release readiness and authorization of replacement software.

SDPS — Software Deployment Standard

Governs deployment, production state, and rollback or recovery.

SCMS — Software Change Management Standard

Governs software changes arising from migration.

SLCS — Software Lifecycle Control Standard

Governs lifecycle state consistency, transition control, exceptions, and

continuity.

SMM may also interact with applicable OOF® identity, authority, accountability, operational reality, data, and autonomous systems governance mechanisms.

These remain independently applicable within their governed spaces.

Practical Applications

Enterprise Application Replacement

Migrating required functionality, users, and data from legacy applications

to successor systems.

Cloud Platform Transition

Transferring services, configurations, interfaces, and operational

dependencies between cloud environments.

Software Supply-Chain Modernization

Replacing unsupported libraries, runtimes, and components while preserving

required behavior.

Data-Intensive Software Migration

Maintaining data integrity, accessibility, identity, and operational

meaning during software replacement.

AI Model and Service Migration

Transitioning applications between AI services while governing

compatibility, behavioral differences, and dependent workflows.

Autonomous Agent Replacement

Transferring tasks and operational responsibilities while independently

validating successor authority and permissions.

Industrial and Embedded Systems

Migrating firmware and software capabilities while preserving hardware

compatibility and physical operational requirements.

Critical Infrastructure

Governing phased transitions, parallel operation, verification, and

recovery where service interruption could have significant consequences.

Practical Use Case 1 — Legacy Financial Software Migration

An organization plans to retire a legacy financial application and migrate

its functions and historical records to a modern platform.

The destination platform successfully imports the source database.

An initial project report therefore classifies the migration as completed.

SMM evaluates:

Source State → Destination Requirements → Data Mapping → Functional

Continuity → User Access → Verification → Actual Outcome

The assessment identifies:

• Most financial records were transferred successfully

• Historical transaction identifiers were transformed

• A required reporting function depends on relationships that were not

• Some archived records remain accessible only through the source

• The destination is operational but does not yet satisfy every required

preserved correctly. application. historical reporting obligation.

The migration classification becomes:

Partially Migrated — Data Relationship and Historical Access Obligations

Outstanding.

SMM requires the unresolved conditions to remain traceable and prevents the

completed data import from being treated as proof of successful migration.

Data Imported ≠ Data Meaning Preserved.

Practical Use Case 2 — Autonomous Agent Replacement

An organization replaces an autonomous software engineering agent with a

newer system.

The new agent can perform the same general development tasks and has passed functional tests.

The organization plans to transfer all ongoing maintenance responsibilities

and retire the original agent.

SMM evaluates:

Source Agent → Successor Agent → Active Tasks → Tool Dependencies → Memory

References → Authority Conditions → Execution Transition → Verification

The assessment identifies:

• The original agent has unfinished maintenance tasks

• The successor requires different repository and deployment permissions

• Some scheduled actions remain linked to the original agent identity

• Historical task records must remain accessible

• The successor's technical capability does not establish authorization

to inherit the original agent's delegated authority.

SMM establishes separate requirements for task continuity, permission

authorization, scheduled activity transition, and verification of the resulting operational state.

The migration classification becomes:

Migration Conditionally Ready — Authority and Task Continuity

Preconditions Outstanding.

Functional Replacement ≠ Authorized Responsibility Transfer.

Critical Distinctions

Migration Planned ≠ Migration Authorized.

Migration Authorized ≠ Every Constituent Action Authorized.

Source Identified ≠ Source State Verified.

Destination Available ≠ Destination Suitable.

Technical Compatibility ≠ Operational Equivalence.

Functional Similarity ≠ Functional Continuity.

Data Imported ≠ Data Meaning Preserved.

Data Transfer Completed ≠ Data Integrity Verified.

User Account Created ≠ User Access Continuity Established.

Interface Available ≠ Interface Compatibility Verified.

Dependency Transferred ≠ Dependency Suitability Established.

Task Continuity ≠ Authority Continuity.

Successor Agent Capable ≠ Successor Agent Authorized.

Migration Executed ≠ Migration Outcome Verified.

Cutover Completed ≠ Source Dependencies Eliminated.

Destination Operational ≠ Source Retirement Authorized.

Parallel Operation ≠ State Consistency.

Recovery Executed ≠ Original State Restored.

Partial Migration ≠ Complete Migration.

Migration Closure ≠ Retirement Closure.

Canonical Boundary

Software Migration Module (SMM) governs the evidence-supported transition

of software functions, data, users, interfaces, configurations, dependencies, and operational responsibilities from identifiable source conditions toward defined destination conditions within the software retirement lifecycle.

It establishes migration identity, scope, source–destination relationships,

transformation requirements, continuity conditions, applicable authority, execution traceability, verification, and outcome classification.

SMM does not independently determine that the source software may be retired merely because migration activities have completed.

It does not replace specialized requirements, software creation, configuration, security, release, deployment, or data governance.

It does not automatically transfer authority between actors, systems, or autonomous agents.

Its principal governance output is the:

Software Migration Governance Profile

This output establishes the evidence-supported relationship between what

was intended to migrate, what actually changed, what was verified, and what remains unresolved.

Canonical Closing Statement

A software migration is not successful merely because a new system is running.

A replacement application may be operational while historical data remains inaccessible.

A database may be transferred while essential relationships lose their meaning.

An interface may be recreated while downstream consumers remain incompatible.

A cloud service may be replaced while critical workflows continue relying on the original environment.

An autonomous agent may acquire the predecessor's tasks without possessing legitimate authority to execute them.

And a software platform may pass technical deployment checks while failing to preserve the operational capabilities it was intended to replace.

SMM establishes the governance structure required to distinguish these

conditions.

It identifies the source and destination as separate operational realities

connected by explicit migration requirements.

It governs what must be preserved, transformed, transferred, replaced, or deliberately discontinued.

It requires the selected migration method and execution pathway to remain

justified and traceable.

It preserves authority boundaries across human, automated, and autonomous

migration activities.

It evaluates the actual destination state against the intended outcome.

And it ensures that unresolved source dependencies and migration obligations remain visible to subsequent retirement governance.

Migration is not the movement of software from one location to another. It is the verified transition of required capabilities and responsibilities between operational states.

Software Migration Module (SMM) transforms software migration from a

technical transfer event into a governed, evidence-supported transition that preserves operational continuity, authority integrity, and the conditions necessary for legitimate software retirement.